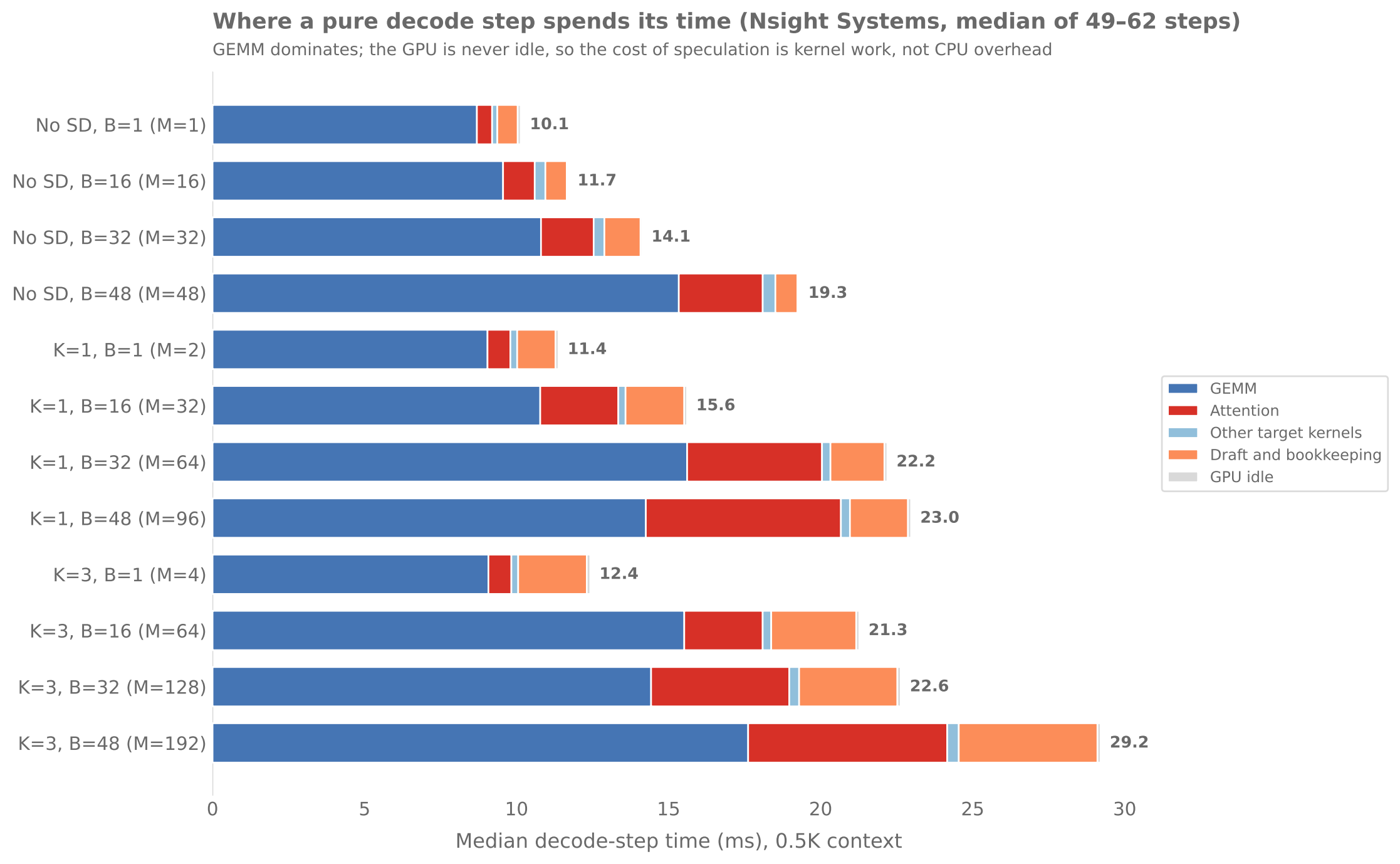
Where a pure decode step spends its time (Nsight Systems, median of 49–62 steps)
GEMM dominates; the GPU is never idle, so the cost of speculation is kernel work, not CPU overhead
No SD, B=1 (M=1)
No SD, B=16 (M=16)
No SD, B=32 (M=32)
No SD, B=48 (M=48)
K=1, B=1 (M=2)
K=1, B=16 (M=32)
K=1, B=32 (M=64)
K=1, B=48 (M=96)
K=3, B=1 (M=4)
K=3, B=16 (M=64)
K=3, B=32 (M=128)
K=3, B=48 (M=192)
10.1
11.7
14.1
19.3
11.4
15.6
22.2
23.0
12.4
21.3
22.6
29.2
GEMM
Attention
Other target kernels
Draft and bookkeeping
GPU idle
0
5
10
15
20
25
30
Median decode-step time (ms), 0.5K context